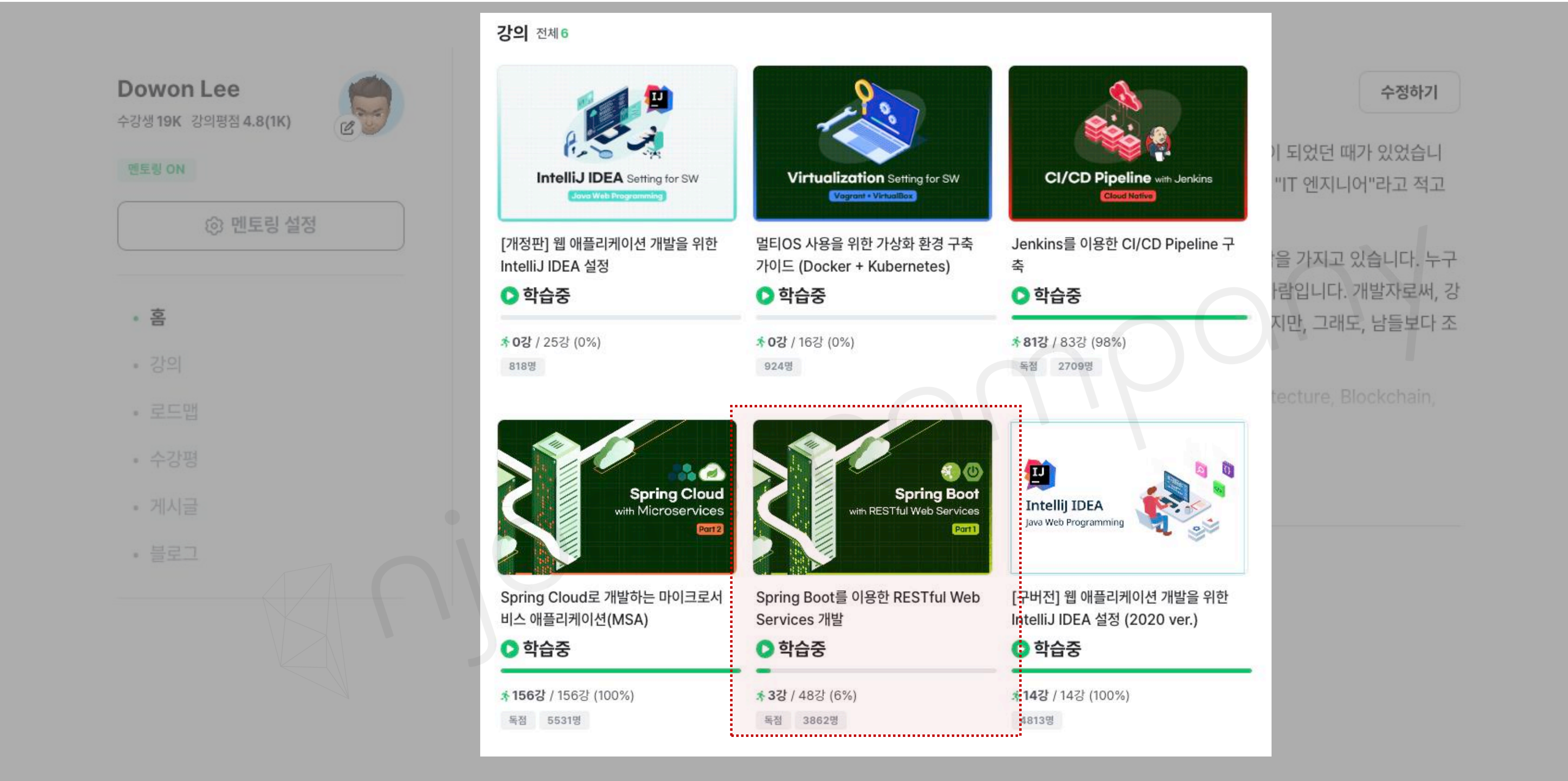
from Beginner to Master

DevOps를 위한 Docker 가상화

Section 8. Logging and Monitoring

프로필



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목차

Section 8.

Logging and Monitoring

- Docker Container Logging
- Docker Container Monitoring

Docker Container Logging

Logging

- 도커 내부에서 발생되는 로그 메시지 확인
  - 도커 실행 시 또는 실행 이후에 발생 되는 로그 메시지 출력

```
$ docker logs <Container ID or Name>
```

  - mariadb 실행에 대한 로그 확인

```
$ docker run -d -p 13306:3306 \
        -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true \
        --name mariadb1 \
        mariadb:latest
```

```
$ docker logs mariadb1
```

Docker Container Logging

Logging (Linux)

- 컨테이너 로그 파일 확인
  - docker inspect → "LogPath": "/var/lib/docker/containers/~
  - /var/lib/docker/containers/${CONTAINER_ID/${CONTAINER_ID}-json.log

```
[ec2-user@ip-172-31-5-111 ~]$ docker ps -a
CONTAINER ID   IMAGE            COMMAND                   CREATED         STATUS         PORTS
7aa484a25939   mariadb:latest   "docker-entrypoint.s…"   7 minutes ago   Up 7 minutes   0.0.0.0:13306->33
```

```
[ec2-user@ip-172-31-5-111 ~]$ sudo ls -al /var/lib/docker/containers/7aa484a2593964038ce8c5e7bb86494bf23f4040293e3ab
total 40
drwx--x--- 4 root root   237 Apr 21 14:29 .
drwx--x--- 3 root root    78 Apr 21 14:29 ..
-rw-r----- 1 root root 14313 Apr 21 14:29 7aa484a2593964038ce8c5e7bb86494bf23f4040293e3ab9ae17de6680c98bab-json.log
```

```
$ sudo cat /var/lib/docker/containers/7aa484a~/7aa484a~-json.log
```

Docker Container Logging

Logging (Linux)

- 컨테이너 로그 파일 → syslog
  - syslog → 유닉스 계열 운영체제에서 로그 수집하는 도구 (커널, 보안, 애플리케이션 로그 등)

```
$ docker run -d -p 13306:3306 --log-driver syslog
            -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true \
            --name mariadb1 \
            mariadb:latest
```

```
[ec2-user@ip-172-31-5-111 ~]$  docker run -d --rm --log-driver=syslog -p 13306:3306 -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true --name mariad
b1 mariadb:latest
c68c243acbb35b9c853e4947b556247857469f36a56c0cd02d8a27bb5366e79c
[ec2-user@ip-172-31-5-111 ~]$ journalctl -u docker.service | grep c68c
Apr 21 14:48:09 ip-172-31-5-111.ap-northeast-2.compute.internal dockerd[3101]: time="2024-04-21T14:48:09.291290636Z" level=info msg="Config
ured log driver does not support reads, enabling local file cache for container logs" container=c68c243acbb35b9c853e4947b556247857469f36a56
c0cd02d8a27bb5366e79c driver=syslog
Apr 21 14:48:09 ip-172-31-5-111.ap-northeast-2.compute.internal c68c243acbb3[3101]: 2024-04-21 14:48:09+00:00 [Note] [Entrypoint]: Entrypoi
nt script for MariaDB Server 1:11.3.2+maria~ubu2204 started.
Apr 21 14:48:09 ip-172-31-5-111.ap-northeast-2.compute.internal c68c243acbb3[3101]: 2024-04-21 14:48:09+00:00 [Warn] [Entrypoint]: /sys/fs/
cgroup/freezer:/docker/c68c243acbb35b9c853e4947b556247857469f36a56c0cd02d8a27bb5366e79c
Apr 21 14:48:09 ip-172-31-5-111.ap-northeast-2.compute.internal c68c243acbb3[3101]: 10:devices:/docker/c68c243acbb35b9c853e4947b55624785746
9f36a56c0cd02d8a27bb5366e79c
```

Docker Container Logging

Logging (Windows)

- 컨테이너 로그 파일 확인
  - docker inspect → "LogPath": "/var/lib/docker/containers/~
  - /var/lib/docker/containers/${CONTAINER_ID/${CONTAINER_ID}-json.log
- Windows에서는 /var/lib/docker/containers/~ 의 내용 확인 어려움
  - /var/lib/docker/container를 Volume mount로 연결한 Linux Container에서 내용 확인 가능

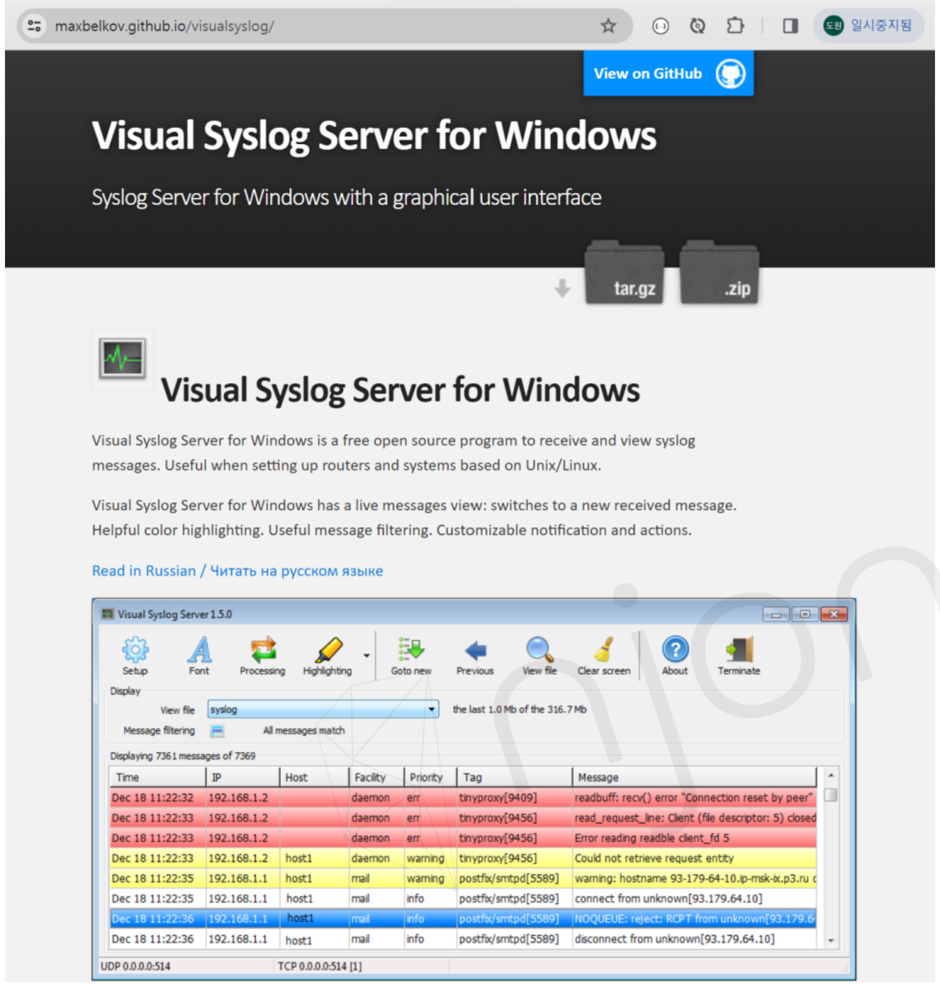
```
C:\> docker run -v /:/data -it ubuntu /bin/bash
# chroot /data
# cd /var/lib/docker/containers
# ls -al
```

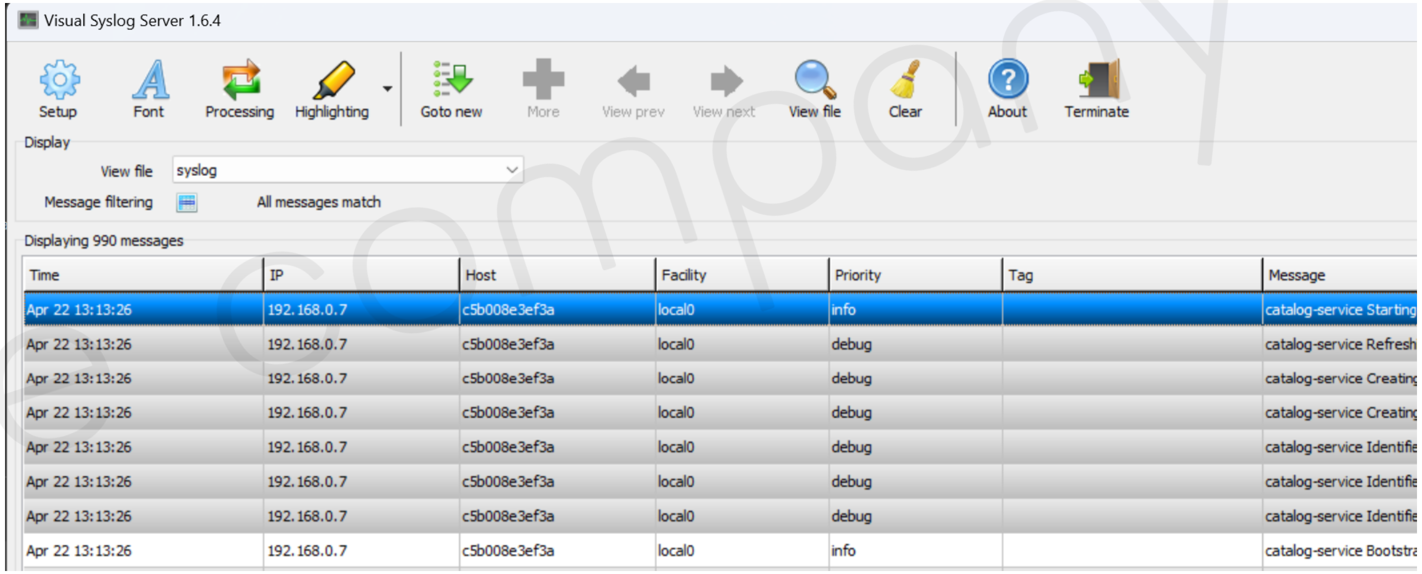
```
drwx--x---  4 root root  4096 Apr 22 14:13 9d19a5067fcbdb80d634b8a5e343d150ec9a7b09aa15d4da9fa291c7f40a769e
drwx--x---  4 root root  4096 Apr 22 14:13 b4526e85340fb2d2fad1cda441b7f332baa5e5bb0a3b1fb13b4ebb0b746ee3ac
drwx--x---  4 root root  4096 Apr 22 14:13 b89558bb32a825b1b317f28b93e51f5c504a3a96edc2d7a2b0202c3c49635826
drwx--x---  4 root root  4096 Apr 22 14:13 b99e9ba87df48130e64d5be668f3747c275534d867fbd61be6c226aa9a855316
drwx--x---  4 root root  4096 Apr 22 14:13 bfef3c20b11c50c4ac418e96c3bf89f0519c2041c50953e8e00172f5f723e218
```

Syslog Server for Windows

Visual Syslog

- https://maxbelkov.github.io/visualsyslog/





Docker Container Logging

Logging (Windows)

▪ *Syslog 서버 설치 필요*

- Windows에서는 Syslog를 사용할 수 없기 때문에, 별도의 Syslog Server 설치

```
$ docker run -d -p 13306:3306 --log-driver syslog \
            --log-opt syslog-address=udp://192.168.0.7:514 \
            -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true \
            --name mariadb1 \
            mariadb:latest
```

| syslog 파일 위치 → C:₩<USER_HOME>₩AppData₩Local₩visualsyslog₩syslog

```
syslog ×
C: > Users > zitan > AppData > Local > visualsyslog > syslog
192.168.0.7   Apr 22 11:54:13   daemon info   c7507dee37c6[1298]   2024-04-22 11:54:13+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.11.2+maria~ubu2204 started.
192.168.0.7   Apr 22 11:54:13   daemon info   c7507dee37c6[1298]   2024-04-22 11:54:13+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
192.168.0.7   Apr 22 11:54:13   daemon info   c7507dee37c6[1298]   2024-04-22 11:54:13+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:10.11.2+maria~ubu2204 started.
192.168.0.7   Apr 22 11:54:13   daemon info   c7507dee37c6[1298]   2024-04-22 11:54:13+00:00 [Note] [Entrypoint]: Initializing database files
192.168.0.7   Apr 22 11:54:13   daemon err    c7507dee37c6[1298]   2024-04-22 11:54:13 0 [Warning] mariadbd: io_uring_queue_init() failed with ENOSYS: check seccomp filters, and the k
192.168.0.7   Apr 22 11:54:13   daemon err    c7507dee37c6[1298]   2024-04-22 11:54:13 0 [Warning] InnoDB: liburing disabled: falling back to innodb_use_native_aio=OFF
192.168.0.7   Apr 22 11:54:16   daemon info   c7507dee37c6[1298]   PLEASE REMEMBER TO SET A PASSWORD FOR THE MariaDB root USER !
192.168.0.7   Apr 22 11:54:16   daemon info   c7507dee37c6[1298]   To do so, start the server, then issue the following command:
192.168.0.7   Apr 22 11:54:16   daemon info   c7507dee37c6[1298]   '/usr/bin/mariadb-secure-installation'
192.168.0.7   Apr 22 11:54:16   daemon info   c7507dee37c6[1298]   which will also give you the option of removing the test
192.168.0.7   Apr 22 11:54:16   daemon info   c7507dee37c6[1298]   databases and anonymous user created by default.  This is
192.168.0.7   Apr 22 11:54:16   daemon info   c7507dee37c6[1298]   strongly recommended for production servers.
192.168.0.7   Apr 22 11:54:16   daemon info   c7507dee37c6[1298]   See the MariaDB Knowledgebase at https://mariadb.com/kb
```

Logging (Windows)

- *Syslog 서버 설치 필요*

- my-backend 예제 사용 → section8-win branch
  - https://github.com/joneconsulting/devops-docker/tree/main/my-backend

```
$ docker-compose —-f docker-compose-win.yml up -d
```

```yaml
version: '3.7'
services:
  my-backend:
    image: edowon0623/catalog-service:syslog-ng-v1.3
    environment:
      - spring.datasource.url=jdbc:mariadb://my-db:3306/mydb
      - spring.datasource.password=
      - DOCKER-COMPOSE-SYSLOG-ADDRESS=192.168.0.7
      - DOCKER-COMPOSE-SYSLOG-PORT=514
    ports:
      - 8088:8088
    depends_on:
      - my-db
    networks:
      - my-network
```

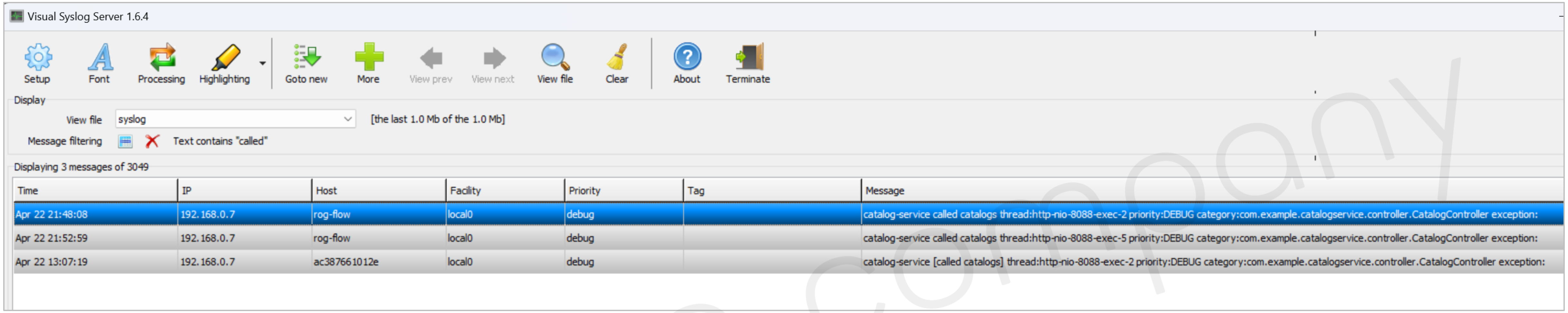
- *Windows PC ip-address*

```xml
<configuration debug="true" scan="true" scanPeriod="30 seconds">
    <property name="SYSLOG-ADDRESS" value="${DOCKER-COMPOSE-SYSLOG-ADDRESS}"/>
    <property name="SYSLOG-PORT" value="${DOCKER-COMPOSE-SYSLOG-PORT}"/>

    <appender name="SYSLOG" class="ch.qos.logback.classic.net.SyslogAppender">
        <syslogHost>${SYSLOG-ADDRESS}</syslogHost>
        <facility>LOCAL0</facility>
        <port>${SYSLOG-PORT}</port>
        <throwableExcluded>true</throwableExcluded>
        <suffixPattern>catalog-service %m thread:%t priority:%p category:%c
        exception:%exception</suffixPattern>
    </appender>
```

Logging (Windows)

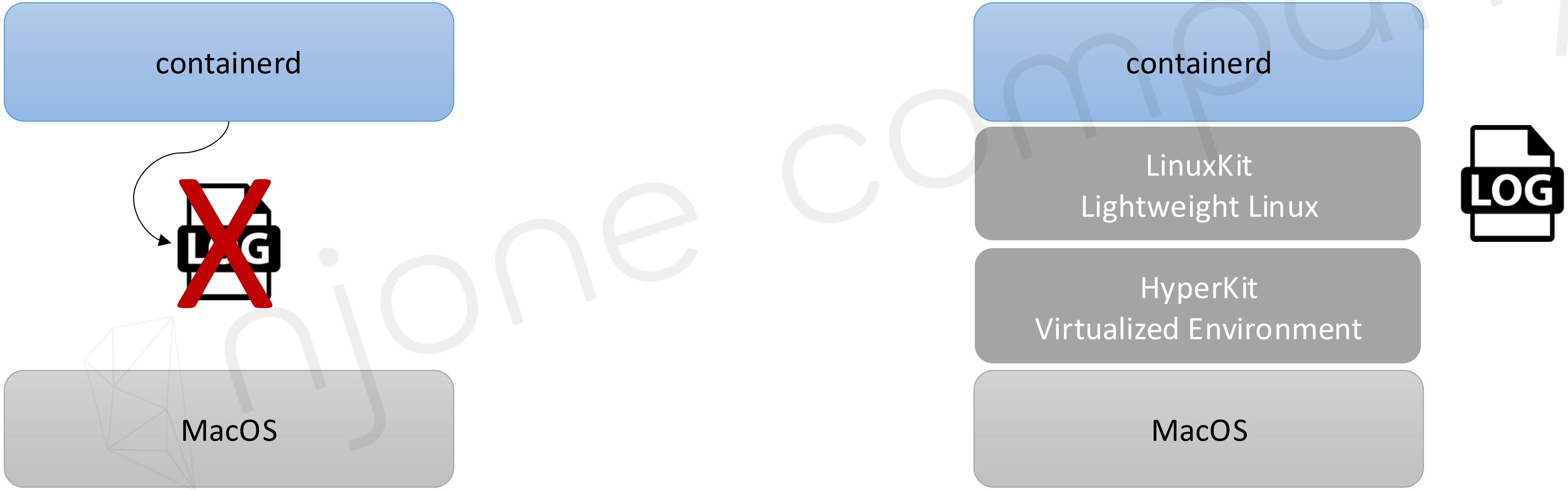
- Spring Boot Application + MariaDB → Syslog



Docker Container Logging

Logging (MacOS)

- 컨테이너 로그 파일 확인
  - docker inspect → "LogPath": "/var/lib/docker/containers/~
  - /var/lib/docker/containers/${CONTAINER_ID/${CONTAINER_ID}-json.log



Docker Container Logging

Logging (MacOS)

- MacOS Apple chip 에서는 /var/lib/docker/containers/~ 의 내용 확인 어려움
  - https://gist.github.com/BretFisher/5e1a0c7bcca4c735e716abf62afad389

```
$ docker run -it --rm --privileged --pid=host justincormack/nsenter1

# cd /var/lib/docker/containers

# ls -al
```

```
docker run -it --rm --privileged --pid=host justincormack/nsenter1
~ # cd /var/lib/docker/containers
/var/lib/docker/containers # ls -al
total 16
drwx--x---    4 root     root          4096 Apr 25 08:55 .
drwx--x---   12 root     root          4096 Apr 25 07:51 ..
drwx--x---    4 root     root          4096 Apr 25 08:41 b8423bbc4892bcaa0dbce80562f8038f4ab0ca4f9f0d23206e875eb0e4368cdf
drwx--x---    4 root     root          4096 Apr 25 08:55 c9fbbc4457fde5b6c3df3b71a13fdf33e7e599c32abaec519d34d503e89a304d
```

Docker Container Logging

Logging (MacOS)

- 컨테이너 로그 파일 → syslog

  | syslog → 유닉스 계열 운영체제에서 로그 수집하는 도구 (커널, 보안, 애플리케이션 로그 등)

  | https://github.com/joneconsulting/devops-docker.git → section8 branch

```
$ cd my-brackend
$ docker-compose -f docker-compose.yml up syslog-ng -d
$ docker-compose ps
```

```yaml
services:
  syslog-ng:
    image: linuxserver/syslog-ng:latest
    container_name: syslog-ng
    environment:
      - PUID=1000
      - PGID=1000
      - TZ=Etc/UTC
    volumes:
      - ./syslog-ng/config:/config
      - ./syslog-ng/log:/var/log #optional
    ports:
      - 514:5514/udp
```

Docker Container Logging

Logging (MacOS)

- Mariadb → Syslog
  - mariadb container에서 syslog-ng container로 로그 출력

```
$ docker run -d -p 13306:3306 --log-driver syslog \
        --log-opt syslog-address=udp://172.17.0.1:514 \
        -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=true \
        --name mariadb1 \
        mariadb:latest
```

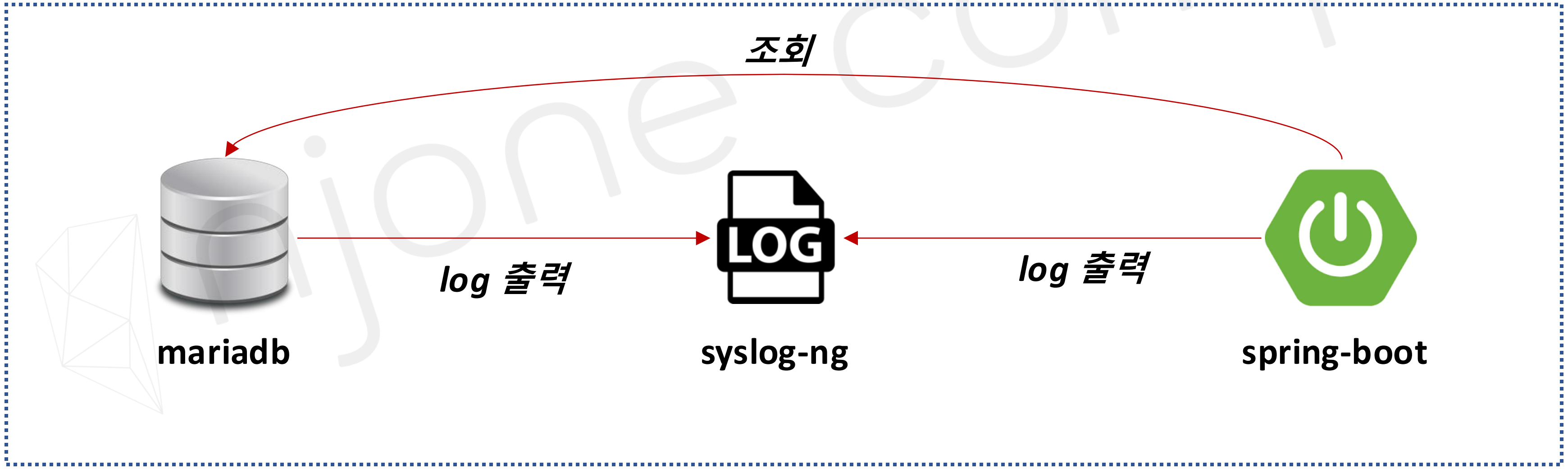
```
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] InnoDB: Buffer pool(s) load completed at 240429 13:30:19
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] Server socket created on IP: '0.0.0.0'.
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] Server socket created on IP: '::'.
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] mariadbd: Event Scheduler: Loaded 0 events
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: 2024-04-29 13:30:19 0 [Note] mariadbd: ready for connections.
Apr 29 13:30:19 172.20.0.1 bd7663fe5b59[311]: Version: '11.3.2-MariaDB-1:11.3.2+maria~ubu2204'  socket: '/run/mysqld/mysqld.sock'  port: 3306  mariadb.org binary distribution
Apr 29 22:30:37 192.168.65.1 AddressBookSourceSync[82513]:
Apr 29 22:30:40 192.168.65.1 AddressBookManager[82546]:
Apr 29 22:30:51 192.168.65.1 AddressBookManager[82612]:
Apr 29 22:31:48 192.168.65.1 powerd[99]: [System: PrevIdle PrevDisp DeclUser kDisp]
```

Logging Demo

- syslog-ng + mariadb + my-backend
  - docker-compose로 syslog-ng / mariadb / my-backend 서비스 실행
  - https://github.com/joneconsulting/devops-docker.git → section8 branch

```
$ cd my-backend

$ docker-compose -f docker-compose-app.yml up -d
```



docker-compose

Docker Container Monitoring

Monitoring

- 일반 Monitoring → 서버의 CPU, 메모리 사용량, 디스크 공간 등의 기본 지표를 추적
- 컨테이너 Monitoring → 컨테이너의 생명 주기가 짧고, 여러 개의 컨테이너 관리 필요
- Prometheus
  - 컨테이너 환경의 애플리케이션을 모니터링 하기 위한 오픈소스 시스템
  - 여러 컨테이너 상태를 개별적으로 추적 (분산 애플리케이션 모니터링 가능)
  - 컨테이너 상태에 대한 Metrics 수집 → 외부 서비스와 연계

Docker Container Monitoring

Prometheus (Linux)

- /etc/docker/daemon.json 파일 수정

```
{
  "metrics-addr": "0.0.0.0:9323",
  "experimental": true
}
```

| Docker 재실행

| Docker Prometheus 실행

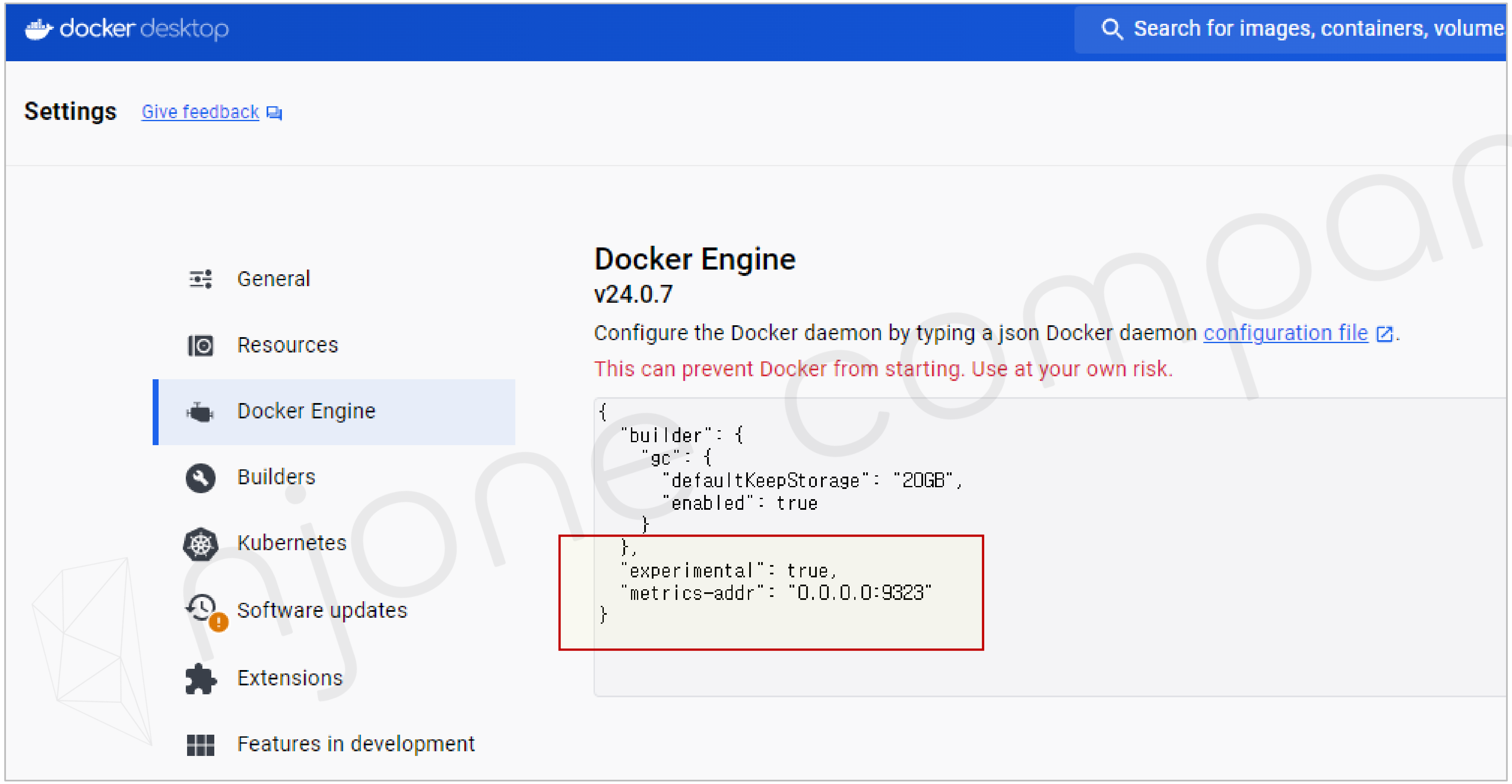
```
$ hostIP=172.31.5.111
```

```
$ docker container run -e DOCKER_HOST=$hostIP -d --rm -p 9090:9090 diamol/prometheus:2.13.1
```

Docker Container Monitoring

Prometheus (Windows & MacOS)

- Docker Desktop > Settings > Docker Engine → Docker Desktop 재실행



Docker Container Monitoring

Prometheus (Windows & MacOS)

- Docker Prometheus 실행

Windows)

```
C:\> set hostIP=[IP Address]
```

```
C:\> docker container run -e DOCKER_HOST=%hostIP% -d --rm -p 9090:9090 diamol/prometheus:2.13.1
```

MacOS)

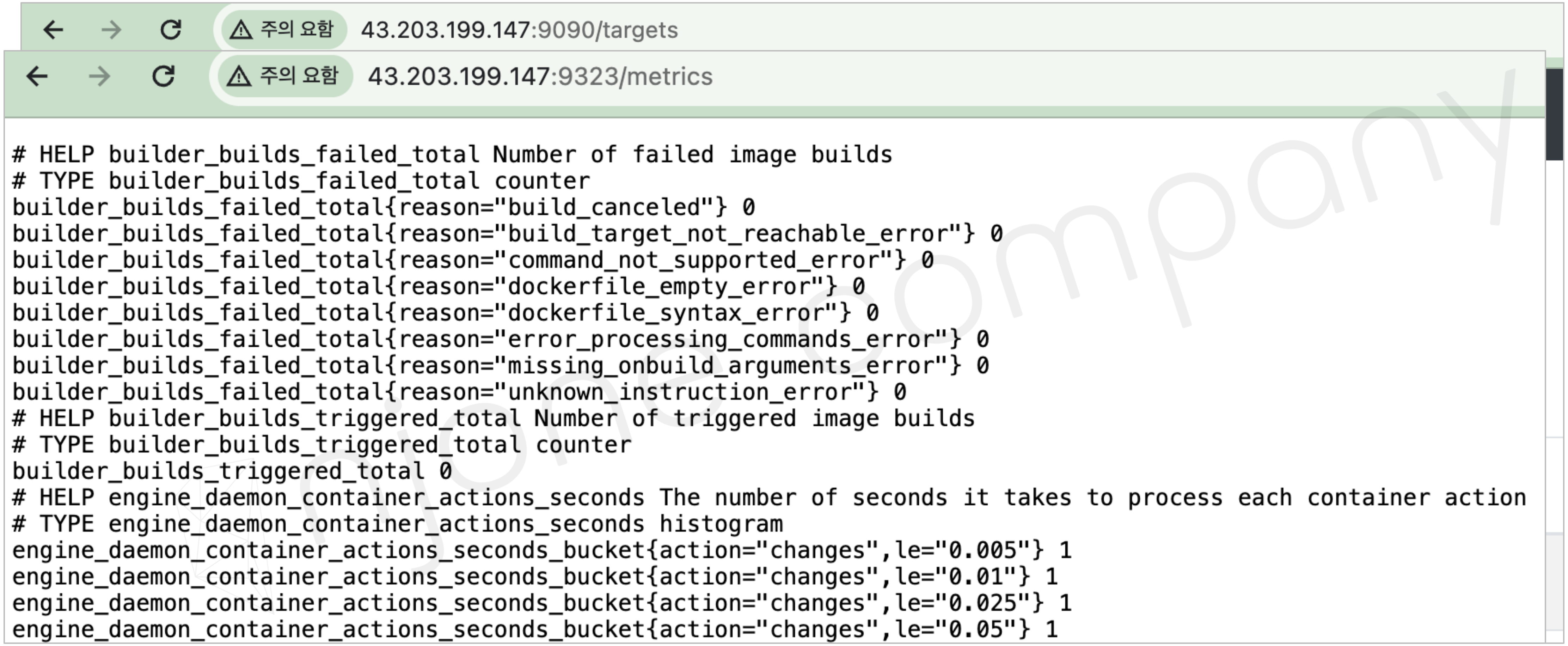
```
$ hostIP=[IP Address]
```

```
$ docker container run -e DOCKER_HOST=$hostIP -d --rm -p 9090:9090 diamol/prometheus:2.13.1
```

Docker Container Monitoring

Prometheus

- Prometheus 실행




```
# HELP builder_builds_failed_total Number of failed image builds
# TYPE builder_builds_failed_total counter
builder_builds_failed_total{reason="build_canceled"} 0
builder_builds_failed_total{reason="build_target_not_reachable_error"} 0
builder_builds_failed_total{reason="command_not_supported_error"} 0
builder_builds_failed_total{reason="dockerfile_empty_error"} 0
builder_builds_failed_total{reason="dockerfile_syntax_error"} 0
builder_builds_failed_total{reason="error_processing_commands_error"} 0
builder_builds_failed_total{reason="missing_onbuild_arguments_error"} 0
builder_builds_failed_total{reason="unknown_instruction_error"} 0
# HELP builder_builds_triggered_total Number of triggered image builds
# TYPE builder_builds_triggered_total counter
builder_builds_triggered_total 0
# HELP engine_daemon_container_actions_seconds The number of seconds it takes to process each container action
# TYPE engine_daemon_container_actions_seconds histogram
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.005"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.01"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.025"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.05"} 1
```

Docker Container Monitoring

Prometheus

- Mariadb 컨테이너 실행

⚠ 주의 요함 43.203.199.147:9323/metrics

```
engine_daemon_container_actions_seconds_bucket{action="start",le="0.25"} 1
engine_daemon_container_actions_seconds_bucket{action="start",le="0.5"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="1"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="2.5"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="5"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="10"} 6
engine_daemon_container_actions_seconds_bucket{action="start",le="+Inf"} 6
engine_daemon_container_actions_seconds_sum{action="start"} 1.513899247000001
engine_daemon_container_actions_seconds_count{action="start"} 6
# HELP engine_daemon_container_states_containers The count of containers in various states
# TYPE engine_daemon_container_states_containers gauge
engine_daemon_container_states_containers{state="paused"} 0
engine_daemon_container_states_containers{state="running"} 2
engine_daemon_container_states_containers{state="stopped"} 0
# HELP engine_daemon_engine_cpus_cpus The number of cpus that the host system of the engine has
# TYPE engine_daemon_engine_cpus_cpus gauge
engine_daemon_engine_cpus_cpus 1
```

Docker Container Monitoring

Prometheus

- Graph